

# Grapevine WGS Pipeline

## SNP-Based Cultivar Identification in *Vitis vinifera*

This pipeline processes paired-end Illumina WGS reads from grapevine cultivar samples through quality trimming, reference alignment, targeted variant calling, and SNP-based cultivar identification using the VIVC (Vitis International Variety Catalogue) database. It is designed to be sample-agnostic and fully reproducible.

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# Pipeline Overview

|                         |                                                          |   |
|-------------------------|----------------------------------------------------------|---|
| Raw FASTQ               | Paired-end Illumina reads                                | ▼ |
| Quality Trimming        | Trimmomatic 0.39 — adapter removal, quality filtering    | ▼ |
| Quality Control         | FastQC — per-base quality, adapter content               | ▼ |
| Reference Mapping       | BWA-MEM2 — alignment to T2T PN40024 reference            | ▼ |
| Variant Calling         | GATK HaplotypeCaller — GVCF mode, 110 SNP positions      | ▼ |
| Genotype Extraction     | bcftools query — CHROM, POS, REF, ALT, GT                | ▼ |
| Excel Processing        | Genotype conversion + strand correction                  | ▼ |
| Cultivar Identification | R script — comparison vs VIVC database (~1874 varieties) |   |

# Requirements

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## System

- Windows 10/11 with WSL2 (Ubuntu)
- Miniconda or Anaconda
- RStudio

## Software Versions

| Tool        | Version | Purpose              |
|-------------|---------|----------------------|
| Trimmomatic | 0.39    | Adapter trimming     |
| FastQC      | 0.12+   | Quality control      |
| BWA-MEM2    | 2.2.1+  | Read alignment       |
| SAMtools    | 1.17+   | BAM processing       |
| GATK        | 4.6.2.0 | Variant calling      |
| BCFtools    | 1.17+   | VCF manipulation     |
| R           | 4.5.1   | Statistical analysis |

## Step 00 — Conda Environment Setup

Create and configure an isolated conda environment for the pipeline.

```
conda create -n wgs_analysis python=3.9
conda activate wgs_analysis

conda config --add channels bioconda
conda config --add channels conda-forge
conda config --set channel_priority strict

conda install -y \
    gatk4 bwa samtools picard trim-galore \
    fastqc bcftools snpeff bedtools vcftools tabix tree

conda install -c bioconda bwa-mem2

gatk --version
```

## Step 01 — Quality Trimming

Trimmomatic 0.39 in paired-end mode. Removes Illumina adapters (TruSeq3-PE), low-quality bases, and reads shorter than 36 bp.

```
BASE="/path/to/wgs_project"
TRIMMOMATIC="/path/to/Trimmomatic-0.39/trimmomatic-0.39.jar"
ADAPTERS="/path/to/Trimmomatic-0.39/adapters/TruSeq3-PE.fa"
SAMPLE="SAMPLE_NAME"

java -jar "$TRIMMOMATIC" PE -threads 4 \
    "$BASE/raw_data/${SAMPLE}_1.fastq" \
    "$BASE/raw_data/${SAMPLE}_2.fastq" \
    "$BASE/trimmed/${SAMPLE}_1_paired.fastq" \
    "$BASE/trimmed/${SAMPLE}_1_unpaired.fastq" \
    "$BASE/trimmed/${SAMPLE}_2_paired.fastq" \
    "$BASE/trimmed/${SAMPLE}_2_unpaired.fastq" \
    ILLUMINACLIP:${ADAPTERS}:2:30:10 \
    LEADING:3 TRAILING:3 SLIDINGWINDOW:4:15 MINLEN:36
```

| Parameter     | Value                 | Description                              |
|---------------|-----------------------|------------------------------------------|
| ILLUMINACLIP  | TruSeq3-PE.fa:2:30:10 | Adapter removal                          |
| LEADING       | 3                     | Remove leading bases below quality 3     |
| TRAILING      | 3                     | Remove trailing bases below quality 3    |
| SLIDINGWINDOW | 4:15                  | Scan with 4-base window, cut if avg < 15 |
| MINLEN        | 36                    | Discard reads shorter than 36 bp         |

## Step 02 — Quality Control

FastQC on trimmed paired reads to assess quality metrics.

```
BASE="/path/to/wgs_project"
SAMPLE="SAMPLE_NAME"

fastqc -o "$BASE/qc" \
  "$BASE/trimmed/${SAMPLE}_1_paired.fastq" \
  "$BASE/trimmed/${SAMPLE}_2_paired.fastq"
```

## Step 03 — Reference Mapping (BWA-MEM2)

BWA-MEM2 alignment to the T2T PN40024 reference genome.

### Index the reference (once only):

```
REF="/path/to/reference/T2T_ref/T2T_ref.fasta"

bwa-mem2 index "$REF"
samtools faidx "$REF"
echo "Reference indexed!"
```

### Map each sample:

```
BASE="/path/to/wgs_project"
REF="$BASE/reference/T2T_ref/T2T_ref.fasta"
SAMPLE="SAMPLE_NAME"

bwa-mem2 mem -t $(nproc) \
  -R "@RG\tID:${SAMPLE}\tSM:${SAMPLE}\tPL:ILLUMINA" \
  "$REF" \
  "$BASE/trimmed/${SAMPLE}_1_paired.fastq" \
  "$BASE/trimmed/${SAMPLE}_2_paired.fastq" | \
samtools view -@ $(nproc) -bS | \
samtools sort -@ $(nproc) -m 4G \
  -o "$BASE/mapping/${SAMPLE}_sorted.bam"

samtools index "$BASE/mapping/${SAMPLE}_sorted.bam"
```

### BAM Quality Control:

```
BAM="$BASE/mapping/${SAMPLE}_sorted.bam"

samtools flagstat "$BAM"
samtools stats "$BAM" > "${BAM%.bam}.stats.txt"
samtools depth -a "$BAM" | \
  awk '{sum+=3; n++;} END {print "Mean coverage: " sum/n "x"}'
```

| Metric | Expected threshold |
|--------|--------------------|
|--------|--------------------|

|                 |                         |
|-----------------|-------------------------|
| Mapped reads    | > 95%                   |
| Properly paired | > 80%                   |
| Mean coverage   | > 15x (30x recommended) |

## Step 04 — Variant Calling (GATK HaplotypeCaller)

GATK HaplotypeCaller in GVCF mode, restricted to the 110 diagnostic SNP positions defined in the interval BED file.

■ Requires *vivc112.bed* — not included in this repository (third-party file).

```
BASE="/path/to/wgs_project"
SAMPLE="SAMPLE_NAME"

gatk HaplotypeCaller \
  -R "$BASE/reference/T2T_ref/T2T_ref.fasta" \
  -I "$BASE/mapping/${SAMPLE}_sorted.bam" \
  -L "$BASE/reference/vivc112.bed" \
  -O "$BASE/vcf_files/${SAMPLE}.g.vcf.gz" \
  -ERC GVCF
```

### Extract genotypes:

```
bcftools query \
  -f '%CHROM\t%POS\t%REF\t%ALT\t[%GT]\n' \
  "$BASE/vcf_files/${SAMPLE}.g.vcf.gz" \
  > "$BASE/vcf_files/${SAMPLE}.txt"
```

Output columns: CHROM | POS | REF | ALT | GT (0/0, 0/1, 1/1)

## Step 05 — Genotype Processing in Excel

Import the .txt genotype file into Excel. Each sample's .xlsx contains two sheets:

| Column         | Description                                   |
|----------------|-----------------------------------------------|
| CHROM          | Chromosome                                    |
| POS            | Position (T2T coordinates)                    |
| REF            | Reference allele                              |
| ALT            | Alternative allele                            |
| GT             | Numeric genotype (0/0, 0/1, 1/1)              |
| GT_L           | Genotype in letters with slash (e.g. A/G)     |
| GT_CORRECT_2   | Strand-corrected genotype, no slash (e.g. AG) |
| SNP_ID         | SNP name from VIVC panel                      |
| SUBJECT_STRAND | Strand orientation (plus or minus)            |

### Excel formulas used:

```

# Convert numeric GT to letter genotype (PT Excel – column GT_L)
=SE(F2="0/0";C2&"/"&C2;SE(F2="0/1";C2&"/"&D2;SE(F2="1/1";D2&"/"&D2;"")))

# English Excel equivalent
=IF(F2="0/0",C2&"/"&C2,IF(F2="0/1",C2&"/"&D2,IF(F2="1/1",D2&"/"&D2,"")))

# Look up SNP name from reference panel
=ÍNDICE(vivc110snps!$A:$A;CORRESP(B2;vivc110snps!$C:$C;0))

# Look up strand
=ÍNDICE(vivc110snps!$D:$D;CORRESP(B2;vivc110snps!$C:$C;0))

```

■ SNPs with *SUBJECT\_STRAND* = minus require complement correction: A↔T, C↔G. Example: GT\_L = T/C on minus strand → GT\_CORRECT\_2 = AG

## Step 06 — VIVC Cultivar Identification (R)

The R script `comparar_amostra_vivc_v4.R` compares each sample's corrected SNP profile against the VIVC database (~1874 varieties) using the 110 diagnostic SNPs.

■ Requires `vivc_snp_db.txt` — not included in this repository (third-party file).

### R packages required:

```
install.packages("readxl")
install.packages("openxlsx")
```

### Configuration — edit only these lines:

```
rm(list = ls()) # clear environment before each run
library(readxl)

FICHEIRO_AMOSTRA <- "/path/to/genotypes/SAMPLE_NAME.xlsx"
NOME_AMOSTRA     <- "SAMPLE_NAME" # must match Excel sheet name
FICHEIRO_DB      <- "/path/to/reference/vivc_snp_db.txt"
PASTA_OUTPUT     <- "/path/to/results_vivc/"
```

Run with Ctrl+Shift+Enter in RStudio.

### Output file — `SAMPLE_NAME_results.txt`:

| Column       | Description                                  |
|--------------|----------------------------------------------|
| score        | Number of SNPs matching the database variety |
| md_count     | Number of SNPs with missing data             |
| pct_md       | Proportion of missing data (md_count / 110)  |
| accession    | Variety name in VIVC                         |
| SNP* columns | TRUE = match, FALSE = mismatch or missing    |

Results are sorted by score descending. A high score with low md\_count indicates a strong cultivar match. A score equal to 110 with md\_count = 0 confirms identity.

## Reference Genome

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The T2T (Telomere-to-Telomere) PN40024 reference genome was used for all mapping and variant calling steps. Available at:

<https://grapedia.org/files-download/>

Shi X. et al. (2023). The complete reference genome for grapevine (*Vitis vinifera* L.) genetics and breeding. Horticulture Research, 10(5), uhad061.

## Third-Party Files (Not Included)

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| File                  | Description                                                           |
|-----------------------|-----------------------------------------------------------------------|
| vivc112.bed           | BED file with 110 SNP positions in T2T coordinates (GATK --intervals) |
| vivc110snpsv5v12x.txt | SNP panel with positions in 12x and T2T coordinates + strand info     |
| vivc_snp_db.txt       | VIVC database with genotypes for ~1874 varieties (110 SNPs)           |

## Notes & Troubleshooting

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- The pipeline runs in WSL2 (Windows Subsystem for Linux 2) with conda.
- Large files (BAMs, FASTQs, GVCFs) are excluded via .gitignore.
- For each new sample, only the SAMPLE variable needs to be updated in the scripts.
- SNP strand correction is critical — incorrect orientation leads to systematically wrong comparisons with the VIVC database.
- The R script must be run with `rm(list = ls())` at the start to avoid variable carry-over between samples in RStudio.
- The reference must be indexed with both `bwa-mem2` index and `samtools faidx` before mapping.
- Running the pipeline on WSL native filesystem (`~/path`) is significantly faster than on the Windows filesystem (`/mnt/c/`).